Diffusion Posterior Sampling for SAR Despeckling

Zelong Wang, Jialing Han, and Chenlin Zhang

***Abstract*— Despeckling is an important preprocessing step for synthetic aperture radar (SAR) image understanding and interpretation, and it can be regarded as an ill-posed inverse problem that heavily depends on the exploration of proper priors on clean images. Motivated by remarkable capability of deep generative priors, we aim to leverage diffusion denoising probabilistic model (DDPM) to achieve a good tradeoff between speckle suppression and detail preservation. To do so, we design a diffusion posterior sampling scheme based on the approximation of likelihood score function via estimation of reverse transition distribution and utilization of multiplicative noise model. The proposed method has two distinct superiorities and the first one lies in the pretrained model, escaping from the limitations of the pairs of clean and speckled images and the high computation complexity of supervised training. More importantly, the proposed method needs no transforms of SAR images, avoiding changing the statistical distribution of speckle and approximating the complex distribution in transform domain. Finally, we empirically validate the proposed method on synthetic and real SAR images, and the visual effects and the quantitative evaluation of the experimental results indicate the superiority of the proposed method over the state-of-the-art compared approaches. Besides, the robustness of the proposed method is also validated via analyzing its performance under the conditions of different speckle intensities, different initial samples (ISs) of diffusion models, and different values of the algorithm parameters.**

***Index Terms*— Despeckling, diffusion model, likelihood score function, posterior sampling, pretrained model.**

I. Introduction

SYNTHETIC aperture radar (SAR) is superior to optical imaging at all-day, all-weather, and strong detection for Earth observations, including oil leakage in the oceans, melting glaciers, landscape classification, disaster monitoring, and surface change detection [1], [2], and [3]. However, the coherent imaging of SAR always results in the multiplicative speckle noise [4], stemming from the fact that the signal backscattered by an SAR resolution cell is the coherent sum of random radar returns from natural randomly rough distributed surface [5], [6]. Speckle noise can degrade SAR image quality seriously, i.e., covering the weak scattering targets and roughening the homogeneous regions, and so it has become one of the most disadvantageous factors for SAR image understanding and interpretation [7], [8].

Mathematically speaking, speckle suppression can be viewed as an ill-posed inverse problem and it aims to restore the clean image from the speckled image by solving the ill-posedness. As we know, exploring proper priors on clean images is important for seeking for the unique solution to the ill-posed inverse problem, thus speckle suppression heavily depends on various priors explored from SAR domain knowledge as well as the data samples [9], [10]. These priors can be categorized into two types, i.e., handcrafted priors and data-driven priors, and we briefly review the related methods for speckle suppression in the viewpoint of inverse problems as follows.

The kernel of traditional approaches for speckle suppression mainly lies in the handcrafted priors, e.g., smoothness, sparseness, as well as their variants. Smoothness, also known as spatial correlation, is one of the earliest priors explored for removing the speckle. This prior can be implemented by means of local or nonlocal strategies, where the former refers to spatial domain filtering techniques such as Lee filter [11], Kuan filter [12], MAP filter [13] to name a few. These methods usually estimate the noise-free pixel values by sliding a window pixel by pixel, but they fail to distinguish noise and edges, resulting in simultaneously smoothing both noise and edges in SAR images. To address this hurdle, many partial differential equations (PDEs) [14], [15] have been designed to preserve the edges when suppressing the speckle. It has been theoretically verified that the Gaussian filtering is equivalent to isotropic diffusion equation, and so various types of diffusion equations, such as nonlinear diffusion equations [16], [17] and anisotropic diffusion PDE [18], have been put forth to well preserve the edges and texture in SAR images. In brief, both the filters and PDEs consider the spatial correlation of adjoint pixels, while the nonlocal approaches [19], [20], [21] associate all of the similar image patches together to achieve a good tradeoff between speckle suppression and edge preservation. Of course, the cost is that the nonlocal methods often have higher computation complexity than the local ones. As the variant of smoothness, similarity test with (context) covariance matrix [22] is also used as speckle filter in the applications of ship detection [23] and polarimetric SAR interferometry [24], achieving good performance.

Besides the smoothness, sparseness is another significant handcrafted prior for despeckling. Sparseness originally stems from the Prony method for sparse signal restoration in 1795 [25], and then is developed in statistics and various applications such as image processing. Specifically, compressive sensing [26], [27] novelly combines sparse recovery and compressive sampling together, greatly promoting the application of sparseness in speckle suppression [28], [29], [30], [31], [32], [33]. Generally speaking, sparse despeckling mainly focus on sparse representation [28], [29] (e.g., wavelets and

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dictionary learning), sparsity measurement [30], [31] (e.g., ℓ_p $(0 \leq p \leq 1)$ norm and various statistical distributions), and sparse recovery algorithms [32], [33] (e.g., Bregman algorithm and Bayesian learning algorithm). These methods are good at preserving details and enhancing targets' structures in SAR images, while speckle noise sometimes cannot be removed completely. A natural idea is to associate the smoothness and the sparseness together, and BM3D [34], [35] just follows this idea by combining nonlocal means filtering and 3-D block matching. Thus BM3D achieves the state-of-the-art performance when it is modified for speckle suppression. For a comprehensive review of SAR despeckling based on handcrafted priors, please refer to [36], [37], and [38] for more details.

As the popularity of deep learning (DL), data-driven priors have been employed to improve the performance of despeckling. To implicitly explore the priors, all sorts of networks have been designed for despeckling and the loss functions have also been optimized. For instance, SAR-CNN [39] and ID-CNN [40] make use of CNN architecture to learn the priors hidden in the training samples, while ID-GAN [41] is based on generative adversarial network equipped with a combination of Euclidean loss, perceptual loss, and adversarial loss. Despite remarkable despeckling capacity, these methods require supervision for training, and it is well-known that clean SAR images are actually impossible. To tackle this issue, various approaches have been proposed. Speckle2Void [42], a self-supervised Bayesian despeckling method, is designed with the training only on noisy SAR images, and the method in [43] utilizes multiobjective neural network trained with generic statistical samples that are constructed using a multicategory generalized Gaussian coherent SAR simulator. Sublook2Sublook [44] is another self-supervised network and it can be trained by the sublook SAR images instead of the clean SAR pairs, where this alternation can be guaranteed by the statistical equivalence between their cost functions. As the attention module arises, MRDDANet [45], SAR-CAM [46], and SAR-ON [47] bring the attention-related structure into the networks to learn multiscale or multilevel priors. Besides, Trans-SAR [48] adopts the popular Transformer architecture and it also has achieved good performance of speckle suppression. More details on DL, e.g., network structures, loss functions, training schemes, experimental results, and so on, can be found in [49], [50], and [51].

Although data-driven priors can be well explored by the methods above, they are always restricted by some limitations. One of them is the training on pairs of clean and speckled images, where the clean SAR images are usually generated by multitemporal fusion or the speckled images are synthesized from clean optical images. Therefore, the generated clean images or the synthetically speckled images may differ from the ground truths, and so the training cannot be guaranteed. Moreover, the training is usually task-specific, so the networks need retraining even if the forward model in the task of despeckling, such as the distributions of speckle, slightly changes. To overcome these difficulties, deep image priors have been explored by untrained neural networks (UNNs) [52], [53], and the basic principle of UNN is that the low-frequency components are usually first learned by the UNN with a start of random weights and random inputs, and the loss functions can be defined as some distances between the outputs and the speckled images. Therefore, stopping the training at the right time will help for restoring the clean images and typical examples include DIP-SR [54], AttentionDIP [55], and NL-DIP [56]. UNN mainly depends on the network models instead of drawing any information from training samples, and so the upper bound of its performance for despeckling has been limited.

Deep generative prior, also known as one of data-driven priors, is usually explored by deep generative networks, such as VAE [57], GAN [58], autoregressive model [59], and normalizing flow [60]. Recently, diffusion models [61], [62], [63], [64] have demonstrated remarkable capability of exploring deep generative priors, and so have become one of the hottest topics in computer version. To date, various diffusion models, discrete or continuous [65], [66], unconditional or conditional [67], [68], and variance-exploding or variance-preserving [69], [70], [71], have been proposed for diverse image generation tasks. For despeckling, several attempts [72], [73] have been made to achieve a good balance between speckle suppression and detail preservation. However, they either need the supervised training on pairs of clean and speckled images or change the distributions of speckle by some transforms. More preciously, the clean SAR images are practically unavailable for supervised training and the noise obtained by popular transforms (e.g., logarithmic transform) on the multiplicative noise usually does not satisfy the Gaussian distributions strictly.

To address the hurdles above, we propose a novel diffusion posterior sampling for SAR despeckling (SAR-DPS), where the speckled images are used as the conditions for reverse denoising process of diffusion models and the deep neural networks can be pretrained in unsupervised mode. To do so, the key is to compute the posterior score function via estimating the likelihood score function. Therefore, we first approximate the reverse transition distribution, and then estimate the likelihood score function from the multiplicative noise model. Finally, the algorithm of SAR-DPS is presented and the experimental results indicate that the proposed method outperforms the state-of-the-art methods for SAR despeckling.

Our contributions are summarized as follows.

1) We present a novel diffusion posterior sampling for SAR despeckling, which only needs the unsupervised pretrained model instead of task-specific supervised training.
2) We design the SAR-DPS algorithm for SAR despeckling, where the statistics of speckle are directly employed without any transforms.
3) We experimentally evaluate the superiority and robustness of SAR-DPS on various SAR images, including synthetic SAR images, TerraSAR-X images, Gaofen-3 images, MiniSAR images, and RADARSAT-1 images.

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II. Related Works

Diffusion models have been widely used for noisy inverse problems [74], [75], [76], [77], such as image deblurring, image inpainting, and image super-resolution. However, their applications for SAR despeckling are just in a fledging period, which stems from the fact that the speckle is multiplicative noise instead of additive noise and the clean SAR images are practically unavailable.

SAR-diffusion denoising probabilistic model (DDPM) [72] is the first attempt to remove the speckle using diffusion models, where optical images and their synthetically speckled images are used to train the networks and a noise predictor conditioned on the speckled image is proposed to iteratively recover the clean image. Nevertheless, it cannot escape from the trap that the speckled images cannot be fully and reasonably simulated due to the diversity of practical speckle and the radar scattering characters of targets. Besides, the supervised training mode of SAR-DDPM is not robust enough, since it needs retraining even if the type or the intensity of speckle slightly changes. In [73], an unsupervised SAR despeckling method based on diffusion models is put forth to solve the difficulties of supervised mode. However, this method starts from the logarithmic transform of speckled images and assumes that the superposition of Gaussian distribution can be used to replace the log Gamma distribution. In fact, the transform has changed the noise distribution and the approximation is relatively rough.

In the viewpoint of inverse problems, SAR images can be regarded as the condition for diffusion models. Different from general-purpose conditional diffusion models, the models for SAR despeckling need specific design to make better use of noise statistics. Assume we have the multiplicative noise model (i.e., the forward model of SAR despeckling) as follows:

$$y = xn \tag{1}$$

where x and y are the corresponding resolution cells on clean SAR image and speckled SAR image, respectively, and n is the speckle satisfying some distributions (e.g., Γ, K, and W distributions). To keep consistent to the denotations in diffusion models, (1) can be rewritten in vector form as follows:

$$\mathbf{y} = \mathbf{x} \odot \mathbf{n} \tag{2}$$

where $\mathbf{x}$ and $\mathbf{y}$ are clean SAR image and speckled SAR image, respectively, $\mathbf{n}$ is the speckle, and $\odot$ denotes the elementwise product. Also, we assume that diffusion models are performed on clean image $\mathbf{x}$ and noise-perturbed image in forward diffusion process is denoted by $\mathbf{x}_t$. Therefore, the speckled SAR image can be used as the guidance of reverse denoising process via diffusion posterior sampling, and the key is to estimate the noise-perturbed conditional distribution $p(\mathbf{y}|\mathbf{x}_t)$ for computing likelihood score function.

The simplest strategy [78] is to ignore the time by replacing $\mathbf{x}_t$ by $\mathbf{x}_0$. Apparently, the approximation is too rough to exactly estimate the likelihood score function for diffusion models. Besides, the noise-perturbed conditional distribution can be rewritten as $p(\mathbf{y}|\mathbf{x}_t) = \int p(\mathbf{y}|\mathbf{x}_0)p(\mathbf{x}_0|\mathbf{x}_t)d\mathbf{x}_0$ from the conditional independence of $\mathbf{y}$ and $\mathbf{x}_t$ with respect to $\mathbf{x}_0$, where $\mathbf{x}_0$ is actually the original clean image $\mathbf{x}$. The two terms in the integration depend on the forward models of inverse problems and the diffusion models, respectively. In case of additive noise model, the first term can be easily derived from the noise distributions and the second term can be estimated by various methods. DMPS [79] directly makes use of uninformative prior of $\mathbf{x}_0$ to approximate $p(\mathbf{x}_0|\mathbf{x}_t)$, which is a simple but biased estimation. DPS [80] presents an accurate estimation of the posterior expectation, and then it simplifies $p(\mathbf{x}_0|\mathbf{x}_t)$ as one-point distribution. Although some works attempt to extend the estimation of $p(\mathbf{y}|\mathbf{x}_t)$ by following DPS or by a heuristic approximation [81], [82], the hyperparameters and the metric functions are difficult to design for further improvements. However, all the methods above are proposed for additive noise with Gaussian or Poisson distributions, and there is no discussion on calculating likelihood score function for multiplicative noise. Therefore, we aim to design a novel diffusion posterior sampling for multiplicative noise and it can be directly applied to SAR despeckling.

III. Diffusion Models

In this section, we will introduce the DDPMs as well as the corresponding continuous versions, providing a basic framework of diffusion posterior sampling for SAR despeckling.

A. Diffusion Denoising Probabilistic Models

DDPM [61] is the first diffusion model for image generation and it consists of a forward diffusion process and a reverse denoising process, which are actually two Markov chains. In the forward diffusion process, the Gaussian white noise is added to the original image iteratively, namely

$$\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\boldsymbol{\varepsilon}_{t-1},\ t = 1, 2, \ldots, T \tag{3}$$

where $\alpha_t = 1 - \beta_t$, $0 < \beta_1 < \beta_2 < \cdots < \beta_T < 1$, $\mathbf{x}_0$ is the original signal, and $\boldsymbol{\varepsilon}_{t-1} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Equivalently, the process in each step can be described as follows:

$$p(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t}\mathbf{x}_{t-1}, (1-\alpha_t)\mathbf{I}),\ t = 1, 2, \ldots, T. \tag{4}$$

According to the recursion in (3) and (4), we can have the distribution of $\mathbf{x}_t$ with respect to the original image $\mathbf{x}_0$. Denote as $\bar{\alpha}_t = \prod_{k=1}^{t} \alpha_k$, we make use of reparameterization trick to obtain

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\varepsilon},\ t = 1, 2, \ldots, T \tag{5}$$

and

$$p(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}),\ t = 1, 2, \ldots, T \tag{6}$$

where $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. With the parameters above, it can be observed that the distribution of $\mathbf{x}_t$ approaches the Gaussian distribution as t increases to T, meaning the real data distribution has been transformed to Gaussian distribution by a Markov chain.

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The reverse denoising process of DDPM can be described as a parameterized Markov chain, namely

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left[\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\right] + (1-\alpha_t)\boldsymbol{\eta}_t \tag{7}$$

for $\boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The unknown parameter θ for noise estimation network $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ in (7) can be learned using variational inference with respect to the reweighted version of the well-known evidence lower bound (ELBO), which is equivalent to optimizing the loss function

$$L_{\text{simple}} = \mathbb{E}_{\mathbf{x}_0,t,\boldsymbol{\epsilon}}\left(\|\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)\|_2^2\right) \tag{8}$$

where $\mathbb{E}$ is the expectation operator and $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. On top of this, the recursion of (7) can be used to refine the samples from T to 0. That is to say, $\mathbf{x}_T$ is first sampled from $\mathcal{N}(\mathbf{0}, \mathbf{I})$, and then $\mathbf{x}_T \to \mathbf{x}_{T-1} \to \cdots \to \mathbf{x}_0$ holds for (7). Therefore, the two Markov chains for forward diffusion process and reverse denoising process can build a bijection between data distribution and Gaussian distribution, so the new sample can be generated via removing the noise from Gaussian distribution by reverse denoising process.

The impressive generation capability of diffusion models results from the deep generative priors explored via Markov chains with deep neural networks. In fact, deep generative priors reflect the real date distribution and can be used to solve the inverse problems of SAR despeckling. Finally, we have to mention that (7) is just a baseline for DDPM, and many improved schemes [62], [63], [64] have been proposed subsequently for high inference speed or precision.

B. Score-Based Diffusion Models

The continuous versions of DDPM are score-based diffusion models [65], [66], providing more freedoms to extend diffusion models and improve their performance. The forward diffusion process of score-based diffusion models can be defined by Itô stochastic differential equation (SDE), i.e.,

$$d\mathbf{x}_t = -\frac{\beta_t}{2}\mathbf{x}_t dt + \sqrt{\beta_t} d\boldsymbol{\omega}, \quad t \in [0, T] \tag{9}$$

where $\beta_t > 0$ is a monotonically increasing function of t and $\boldsymbol{\omega}$ is a standard Wiener process. It has been verified that DDPM is just one special discretization, i.e., the ancestral sampler, from Itô SDE. To date, various advanced sampling methods, e.g., predictor–corrector sampler, have been proposed.

The reverse SDE corresponding to (9) can be written in the form of

$$d\mathbf{x}_t = \left[-\frac{\beta_t}{2}\mathbf{x}_t - \beta_t \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)\right]dt + \sqrt{\beta_t} d\bar{\boldsymbol{\omega}} \tag{10}$$

for $t \in [0, T]$. Denotations dt and $d\bar{\boldsymbol{\omega}}$ mean the infinitesimal negative time step and the standard Wiener process running backward in time, respectively, and $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)$ is the unknown time-dependent score function. In score-based diffusion models, the score function can be also approximated by a score matching network $\mathbf{s}_\theta(\mathbf{x}_t, t)$ and this network can be trained by denoising score matching, which is equivalent to (8). And then, the recursion in (10) can also be utilized to refine the samples from T to 0.

It can been concluded that, by proper setting of the parameters, $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ for the noise $\boldsymbol{\epsilon}$ in DDPM and $\mathbf{s}_\theta(\mathbf{x}_t, t)$ for the score function $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)$ in score-based diffusion models are equivalent to each other, i.e.,

$$-\frac{1}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) = \mathbf{s}_\theta(\mathbf{x}_t, t) \tag{11}$$

providing more freedoms for designing diffusion models. The proof of the statement above can be referred to Appendix.

C. Conditional Diffusion Models

Both the DDPM and the score-based DMs are unconditional diffusion models, meaning that the images are generated randomly by the reverse denoising process and the only requirement for these images is to maintain the same distribution as the training samples, that is, to comply with the deep generative priors explored by the DMs. Therefore, applying unconditional DMs to inverse problems proves challenging due to the difficulty in ensuring that the generated images closely approximate the ground truths. In this case, conditional DMs are proposed to guide the reverse denoising process by the measurement of inverse problems, and thus their basic principle is to replace the score function in (10) by the posterior score function, namely

$$d\mathbf{x}_t = \left[-\frac{\beta_t}{2}\mathbf{x}_t - \beta_t \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t|\mathbf{y})\right]dt + \sqrt{\beta_t} d\bar{\boldsymbol{\omega}} \tag{12}$$

where $\mathbf{y}$ is the measurement and $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t|\mathbf{y})$ is the posterior score function. If the posterior score function can be estimated, we can leverage (12) to implement the guidance for diffusion models. However, it seems difficult as the network in DDPM or score-based DMs can only approximate the score function $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)$. At this point, we can utilize Bayes' rule to factorize the posterior score function as follows:

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t|\mathbf{y}) = \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) + \nabla_{\mathbf{x}_t} \log p_t(\mathbf{y}|\mathbf{x}_t) \tag{13}$$

where the two terms on the right are named the prior score function and the noise-perturbed likelihood score function, respectively. In fact, the prior score function is just the score function in (10) and can be estimated by the trained neural networks $\mathbf{s}_{\hat{\theta}}(\mathbf{x}_t, t)$ in (11). In summary, if the likelihood score function $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y}|\mathbf{x}_t)$ can be well approximated, the posterior sampling scheme can be performed for inverse problems.

Based on the analysis above, we intend to show two fundamental remarks on conditional DMs for inverse problems. On the one hand, the prior score function in conditional DMs is the same as the score function in unconditional DMs, both of which stand for the deep generative priors explored by deep networks of DMs. Moreover, the prior score function has no relations with the forward models of inverse problems, i.e., the multiplicative noise model, and the only requirement is that the prior represented by this function complies with the ground truth of inverse problems. That is to say, the training of the networks for the prior score function in conditional DMs is the same to that in unconditional DMs, and thus the conditional DMs for inverse problems need no specific design of the networks as well as the inference for prior score

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function. On the other hand, the likelihood score function is mainly determined by the networks of DMs as well as the forward models of inverse problems, where the latter only plays the role of guiding the reverse denoising process and has no impact on the forward diffusion process, the training of the networks, and the inference of the prior score function in conditional DMs.

Specifically, when the inverse problem of SAR despeckling is considered within the framework of conditional DMs, the forward model in (2) for SAR despeckling only has impact on the guidance of the reverse diffusion process by the measurement (i.e., the SAR image), rather than the forward diffusion process and the inference of prior score function. This means the Gaussian noise is still added to the samples iteratively to train the networks in the forward diffusion process of conditional DMs for SAR despeckling, and then the prior score function is estimated by the trained networks in the reverse denoising process. Besides, the likelihood score function in the reverse denoising process of conditional DMs for SAR despeckling depends on both the trained networks and the forward model in (2). For example, the multiplicative characteristic and the specific distribution of the noise in (2) only affects the estimation of the likelihood score function, and how we shall approximate this function from the trained networks and the forward model is at the core of SAR despeckling within the framework of conditional DMs.

IV. Diffusion Posterior Sampling

In this section, we propose the diffusion posterior sampling for SAR despeckling, i.e., the SAR-DPS, which is also within the framework of conditional DMs. As we have mentioned above, the forward model in (2) only has impact on the reverse denoising process, and so the forward diffusion process, the training of the network, and the estimation of the prior score function by the trained network still follow the setup of unconditional DMs. Consequently, the key of the SAR-DPS is to estimate the likelihood score function $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{y}|\mathbf{x}_t)$, and the subscript t of p_t will be omitted for short. According to the forward model (2) of SAR despeckling and the forward diffusion process of diffusion models, it can be observed that the speckled image $\mathbf{y}$ and the noise-perturbed image $\mathbf{x}_t$ are independent to each other with respect to the original clean image $\mathbf{x}_0$ [i.e., $\mathbf{x}$ in (2)], so the noise-perturbed conditional distribution $p(\mathbf{y}|\mathbf{x}_t)$ can be rewritten as follows:

$$p(\mathbf{y}|\mathbf{x}_t) = \int p(\mathbf{y}|\mathbf{x}_0)p(\mathbf{x}_0|\mathbf{x}_t)d\mathbf{x}_0. \tag{14}$$

Next, we first estimate the conditional distribution $p(\mathbf{x}_0|\mathbf{x}_t)$, which is just the reverse transition distribution. And then, associating with the conditional distribution $p(\mathbf{y}|\mathbf{x}_0)$ from the forward model (2), we approximate the likelihood score function. Finally, the algorithm of SAR-DPS is designed and its superiorities are also analyzed.

A. Estimation of Reverse Transition Distribution

Reverse transition distribution $p(\mathbf{x}_0|\mathbf{x}_t)$ in (14) is hard to estimate, although the corresponding forward transition distribution $p(\mathbf{x}_t|\mathbf{x}_0)$ can be easily obtained by (6). To address this difficulty, we first introduce the Tweedie's formula as follows.

Theorem 1 (Tweedie's Formula): Suppose $p(\mathbf{z}|\boldsymbol{\eta})$ is a distribution of exponential family, i.e.,

$$p(\mathbf{z}|\boldsymbol{\eta}) = p_0(\mathbf{z}) \exp(\boldsymbol{\eta}^T T(\mathbf{z}) - \varphi(\boldsymbol{\eta})) \tag{15}$$

where $\boldsymbol{\eta}$ is a natural or canonical vector of the family, $\varphi(\boldsymbol{\eta})$ is a cumulant generating function that normalizes the density, $T(\mathbf{z})$ is some function of $\mathbf{z}$, and $p_0(\mathbf{z})$ is the density of $\mathbf{z}$ for $\boldsymbol{\eta} = \eta$. Then the posterior expectation $\mathbb{E}(\boldsymbol{\eta}|\mathbf{z})$ satisfies

$$(\nabla_{\mathbf{z}} T(\mathbf{z}))^T \mathbb{E}(\boldsymbol{\eta}|\mathbf{z}) = \nabla_{\mathbf{z}} \log p(\mathbf{z}) - \nabla_{\mathbf{z}} \log p_0(\mathbf{z}). \tag{16}$$

The proof can be referred to [83]. According to (6), we have

$$p(\mathbf{x}_t|\mathbf{x}_0) = \frac{1}{\left[2\pi(1-\bar{\alpha}_t)\right]^{d/2}} \exp\left(-\frac{\|\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0\|_2^2}{2(1-\bar{\alpha}_t)}\right) \tag{17}$$

where d is the dimension of $\mathbf{x}_0$. The Gaussian distribution in (17) can be rewritten as the canonical decomposition, i.e.,

$$p(\mathbf{x}_t|\mathbf{x}_0) = p_0(\mathbf{x}_t) \exp(\mathbf{x}_0^T T(\mathbf{x}_t) - \varphi(\mathbf{x}_0)) \tag{18}$$

where

$$\begin{cases} p_0(\mathbf{x}_t) = \dfrac{1}{\left[2\pi(1-\bar{\alpha}_t)\right]^{d/2}} \exp\left(-\dfrac{\|\mathbf{x}_t\|_2^2}{2(1-\bar{\alpha}_t)}\right) \\ T(\mathbf{x}_t) = \dfrac{\sqrt{\bar{\alpha}_t}}{1-\bar{\alpha}_t}\mathbf{x}_t \\ \varphi(\mathbf{x}_0) = \dfrac{\bar{\alpha}_t\|\mathbf{x}_0\|_2^2}{2(1-\bar{\alpha}_t)}. \end{cases} \tag{19}$$

Then, by Theorem 1 it is easy to have the following corollary.

Corollary 1: For the DDPM with the forward transition distribution $p(\mathbf{x}_t|\mathbf{x}_0)$ in (6), the reverse transition distribution $p(\mathbf{x}_0|\mathbf{x}_t)$ has the posterior expectation, i.e.,

$$\mathbb{E}(\mathbf{x}_0|\mathbf{x}_t) = \frac{1}{\sqrt{\bar{\alpha}_t}}\left[\mathbf{x}_t + (1-\bar{\alpha}_t)\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)\right]. \tag{20}$$

We have mentioned that the prior score $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)$ can be estimated by the deep neural network $\mathbf{s}_\theta(\mathbf{x}_t, t)$, or equivalently $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$, so, according to corollary 1, we can have the estimation of the posterior expectation

$$\mathbb{E}(\mathbf{x}_0|\mathbf{x}_t) = \frac{1}{\sqrt{\bar{\alpha}_t}}\left[\mathbf{x}_t + (1-\bar{\alpha}_t)\mathbf{s}_\theta(\mathbf{x}_t, t)\right]. \tag{21}$$

Therefore, the reverse transition distribution can be approximated by one-point distribution, i.e.,

$$p(\mathbf{x}_0|\mathbf{x}_t) = \delta(\mathbf{x}_0 - \hat{\mathbf{x}}_0) \tag{22}$$

where $\hat{\mathbf{x}}_0 = \mathbb{E}(\mathbf{x}_0|\mathbf{x}_t)$. The approximation in (22) means the posterior expectation in (21) is used to estimate $\mathbf{x}_0$ from $\mathbf{x}_t$. It is worth noting that the posterior covariance has been ignored when estimating $p(\mathbf{x}_0|\mathbf{x}_t)$, since the posterior covariance approaches zero as $t \to T$ and this omission will provide convenience for computing the likelihood score function in Section IV-B.

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B. Approximation of Likelihood Score Function

To compute the likelihood score function $\nabla_{\mathbf{x}_t} \log p(\mathbf{y}|\mathbf{x}_t)$, we need to further estimate the distribution $p(\mathbf{y}|\mathbf{x}_0)$ in (14), depending on the forward model of inverse problems. For SAR despeckling, its forward model, i.e., the multiplicative noise model, has been presented in (2). At this point, we take the Γ distribution for the example, to show the idea of directly employing the noise statistics to guide the reverse denoising process of diffusion models. The other noise statistics, such as K and W distributions, can be also used in diffusion posterior sampling similarly.

Assume the noise in (2) satisfies the Γ distribution, i.e., $\mathbf{n} \sim \Gamma(L, L/\alpha)$ with the probability density function

$$p(x; L, L/\alpha) = \frac{L^L}{\alpha\Gamma(L)}\left(\frac{x}{\alpha}\right)^{L-1}\exp\left(-\frac{L}{\alpha}x\right) \tag{23}$$

where $\Gamma(\cdot)$ is the Γ function, $x > 0$, and $L > 0$ and α denote the number of looks and the underlying reflectivity mean, respectively. Generally speaking, it is usually assumed $\alpha = 1$, then we have the noise distribution as follows:

$$p(x; L, L) = \frac{L^L}{\Gamma(L)}x^{L-1}\exp(-Lx) \tag{24}$$

where $x > 0$. According to the forward model in (2), the conditional distribution of resolution cell $\mathbf{y}^i$ with respect to $\mathbf{x}_0^i$ can be written as follows:

$$p(\mathbf{y}^i|\mathbf{x}_0^i) = \frac{L^L}{\Gamma(L)}\frac{(\mathbf{y}^i)^{L-1}}{(\mathbf{x}_0^i)^L}\exp\left(-L\frac{\mathbf{y}^i}{\mathbf{x}_0^i}\right) \tag{25}$$

where $\mathbf{x}_0^i \neq 0$ and $i = 1, 2, \ldots, d$. Thus, we can further obtain the joint conditional distribution of $\mathbf{y}$ with respect to $\mathbf{x}_0$, i.e.,

$$p(\mathbf{y}|\mathbf{x}_0) = C_0\frac{(Q(\mathbf{y}))^{L-1}}{(Q(\mathbf{x}_0))^L}\exp\left(-L\sum_{i=1}^{d}\frac{\mathbf{y}^i}{\mathbf{x}_0^i}\right) \tag{26}$$

where $\mathbf{x}_0 \neq \mathbf{0}$, $C_0 = (L^L/\Gamma(L))^d$, and $Q(\cdot)$ is defined as $Q(\mathbf{z}) = \prod_{i=1}^{d}\mathbf{z}^i$. Until now, we have obtained the expression of $p(\mathbf{y}|\mathbf{x}_0)$. Notice that $p(\mathbf{y}|\mathbf{x}_0)$ uncovers the speckle statistics and is directly estimated from the forward model of SAR despeckling, so it does not depend on any transforms of SAR images. Moreover, this approach can be easily extended to other statistics, such as K and W distributions.

Associating with (14), (22), and (26), we have

$$p(\mathbf{y}|\mathbf{x}_t) = C_0\frac{(Q(\mathbf{y}))^{L-1}}{(Q(\hat{\mathbf{x}}_0))^L}\exp\left(-L\sum_{i=1}^{d}\frac{\mathbf{y}^i}{\hat{\mathbf{x}}_0^i}\right). \tag{27}$$

Let $R = \log p(\mathbf{y}|\mathbf{x}_t)$, then the likelihood score function can be rewritten as follows:

$$\nabla_{\mathbf{x}_t}\log p(\mathbf{y}|\mathbf{x}_t) = \frac{\partial\hat{\mathbf{x}}_0}{\partial\mathbf{x}_t}\frac{\partial R}{\partial\hat{\mathbf{x}}_0}. \tag{28}$$

According to (27), it is easy to obtain

$$\frac{\partial R}{\partial\hat{\mathbf{x}}_0} = \frac{L}{\hat{\mathbf{x}}_0 \odot \hat{\mathbf{x}}_0}\odot(\mathbf{y} - \hat{\mathbf{x}}_0) \tag{29}$$

where $\odot$ is the elementwise product. From the definition of $\hat{\mathbf{x}}_0$ in (21), we can have the derivative of $\hat{\mathbf{x}}_0$ with respect to $\mathbf{x}_t$, i.e.,

$$\frac{\partial\hat{\mathbf{x}}_0}{\partial\mathbf{x}_t} = \frac{1}{\sqrt{\bar{\alpha}_t}}\left[\mathbf{I} + (1-\bar{\alpha}_t)\mathcal{J}_{\mathbf{x}_t}(\mathbf{s}_\theta(\mathbf{x}_t, t))\right] \tag{30}$$

where $\mathbf{I}$ is the unit matrix and $\mathcal{J}_{\mathbf{x}_t}(\cdot)$ is the Jacobian operator performing on vector-value functions with respect to $\mathbf{x}_t$. Therefore, we can get the expression of $\nabla_{\mathbf{x}_t}\log p(\mathbf{y}|\mathbf{x}_t)$ by substituting (29) and (30) into (28).

In the derivation of likelihood score function, we note that $\hat{\mathbf{x}}_0$ in (27) should be a nonzero vector for the nonzero vector $\mathbf{x}_0$ in (26). However, it is possible to have zero elements of $\hat{\mathbf{x}}_0$ in diffusion models, so a small positive number can be added to the denominator in (29) to solve this trouble in computing likelihood score function. In this case, the approximation of likelihood score function holds for any values of $\hat{\mathbf{x}}_0$ in diffusion models.

C. SAR-DPS

First, we intend to discuss the parameter L for the posterior score function. This parameter is actually the number of looks, which can be readily obtained from the processing procedures of SAR images. For instance, $L = 1$ holds for SAR images without multilook processing, otherwise the parameter L should be determined according to the multilook processing. When the SAR images for despeckling are with no information on their processing procedures, the parameter L is unknown and here we can estimate it by the equivalent number of looks (ENLs) [84]. Despite their difference, the estimation is robust for despeckling, and so we assume L as a known parameter for given SAR images in the following analysis.

Based on the approximation of posterior score function and the principle of diffusion models, the SAR-DPS is summarized in Algorithm 1. It can be observed that the input of Algorithm 1 includes the trained score matching network $\mathbf{s}_{\hat{\theta}}(\cdot,\cdot)$, whose structure follows the U-Net in [86]. Moreover, we intend to utilize the score matching network pretrained on clean optical images rather than trained from scratch for SAR-DPS, and the reasons primarily lie in two aspects. On the one hand, SAR-DPS is an unsupervised model and the training of the U-Net needs clean SAR images, but real SAR images do not have the corresponding clean ones. In this case, the training samples for the U-Net in SAR-DPS are unavailable. Although the clean images of SAR images can be replaced by the multitemporal SAR images, they may be difficult to acquire or fused accurately and so the feasibility of training from scratch cannot be guaranteed for SAR-DPS. On the other hand, the U-Net can capture the deep generative prior of optical images, if this network is pretrained on clean optical images. Consequently, the pretrained U-Net can be associated with the guidance by SAR images, providing information fusion from SAR images and optical images in the reverse denoising process. Besides, the utilization of the pretrained U-Net also helps for escaping from the high computation cost of training on large-size samples.

Finally, some remarks on this algorithm are listed below. First, the parameter γ in Algorithm 1 is actually the weight of posterior score function, directly determining the guidance intensity of the measurement. In Section V, experimental results would validate the robustness of SAR-DPS with respect to the parameter γ. Second, in the implementation of Algorithm 1, we sequentially perform the prior score function

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Algorithm 1 SAR-DPS

Input: $\mathbf{y}$, $\{\beta_t\}_{t=0}^{T}$, $\mathbf{s}_{\hat{\theta}}(\cdot,\cdot)$, γ and $\{\widetilde{\sigma}_t\}_{t=1}^{T}$
Output: $\mathbf{x}_0$
1: **Initialization:** $\alpha_t = 1-\beta_t$, $\bar{\alpha}_t = \prod_{k=1}^{t}\alpha_k$ and $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0},\mathbf{I})$
2: **for** $t = T$ **to** 1 **do**
3: $\quad\boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0},\mathbf{I})$
4: $\quad\widetilde{\mathbf{x}}_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left[\mathbf{x}_t + (1-\alpha_t)\mathbf{s}_{\hat{\theta}}(\mathbf{x}_t,t)\right] + \widetilde{\sigma}_t\boldsymbol{\eta}_t$
5: $\quad$Calculate $\nabla_{\mathbf{x}_t}\log p(\mathbf{y}|\mathbf{x}_t)$ by (28)
6: $\quad\mathbf{x}_{t-1} = \widetilde{\mathbf{x}}_{t-1} + \gamma\frac{1-\alpha_t}{\sqrt{\alpha_t}}\nabla_{\mathbf{x}_t}\log p(\mathbf{y}|\mathbf{x}_t)$
7: **end for**

and the likelihood score function on the noise-perturbed image, rather than computing the posterior score function directly. Thus we can clearly see the despeckling mechanism of SAR-DPS, namely, the update by prior score function depends on deep generative prior explored by the network $\mathbf{s}_{\hat{\theta}}(\cdot,\cdot)$ and the update by likelihood score function mainly comes from the speckled image $\mathbf{y}$. This mechanism also offers the superiority of SAR-DPS in fusing all kinds of information via pretrained networks. Finally, the method of SAR-DPS can be applied to various diffusion models and we only take that in [86] for example. Therefore, unless specifically stated, we will adopt the same setup of the parameters in Algorithm 1 as that in [86]. For example, we set $T = 1000$ and choose the descending time stepping from $t = T$ to $t = 1$ with a uniformly spaced time-step schedule. Moreover, we adopt linear schedule for β_t from 0.0001 to 0.02 for 1000 timesteps. Even so, some minor improvements are suggested in Algorithm 1. For instance, we replace $1-\alpha_t$ by a learned or given $\widetilde{\sigma}_t$ [61], [63] and also add a step-length parameter γ to the posterior score function to keep a good balance between the two terms.

In the end, we aim to analyze the superiority of SAR-DPS to the existing methods based on diffusion models. SAR-DPS is an unsupervised training, and so no pairs of clean and speckled images are needed. More importantly, the pretrained networks can work well for SAR-DPS, not only significantly improving the robustness of training but also reducing the training time and cost. To some extent, the pretrained model can associate the priors from the training images with the speckled images, even if the training images and the speckled images belong to different image categories. The output of SAR-DPS can be viewed as the fusion of the training images and the speckled images, providing an explanation for the following question: Why can the pretrained networks trained on optical datasets work well for SAR despeckling in the following experiments? Besides, the diffusion models of SAR-DPS always perform in SAR image domain instead of its transform domain, thus we have no loss on the model precision of SAR speckle and need no approximation of noise distributions.

V. Experiments

In order to verify the performance of the proposed method, despeckling experiments on synthetic SAR images and real SAR images are performed. We first present the experimental setup, and then show the experimental results of different SAR images. Finally, we also analyze the robustness of the proposed method with respect to speckle intensity, initial sample (IS) of diffusion model, and the algorithm parameter.

A. Experimental Setup

1) SAR Images: For testing SAR images, we select synthetic SAR images and real SAR images to validate the performance of the proposed method SAR-DPS sufficiently. Synthetic SAR images are obtained by adding the noise satisfying Γ distribution to clean optical remote sensing images from DSIFN-CD dataset [85] via (2). Real single-look SAR images are from the sensors of TerraSAR-X, Gaofen-3, MiniSAR, and RADARSAT-1, and the details of these images will be introduced in Sections V-C–V-F. The dataset of training SAR images consists of 3.6k pairs of clean optical remote sensing images from DSIFN-CD dataset [85] and the corresponding speckled images obtained by adding the noise satisfying Γ distribution with $L = 1$ via (2). In the experiments, all of the despeckling methods that need training are trained on this dataset for fair of comparison. All of the testing and training SAR images in the experiments are single channel and with the size of 256×256.

2) Task: To validate the superiority of SAR-DPS, both synthetic SAR images and real SAR images are used for comparing the despeckling results of the proposed method and the compared methods. However, we only select some representative SAR images, i.e., the synthetic SAR images and the TerraSAR images, to analyze the robustness of the proposed method.

3) Pretrained Model: We have mentioned that SAR-DPS is an unsupervised DL and the pretrained model can be directly leveraged to avoid the costly training. At this point, we select the pretrained diffusion model from [86] for SAR-DPS, since the model architecture has been greatly improved and the pretrained model has been verified to beat generative adversarial networks on image synthesis task [86]. More importantly, the pretrained model has been validated sufficiently in various inverse problems [79], [80], and it can be directly utilized without any fine-tuning for specific tasks. As we have mentioned above, this model is pretrained on the ImageNet dataset, a set of optical natural images, but it can work well for SAR despeckling, since diffusion models indicate that their generated images can be viewed as the information fusion from the training samples (i.e., the optical images) and the guidance of the measurements (i.e., the SAR images). By the way, it is also worth to implement the other pretrained models that have been validated in future works, but now, restricted by computation resource, we only take the pretrained model from [86] for example and do not try the other models.

4) Compared Methods: For all these algorithms, if not stated otherwise, the free parameters are set as suggested in the reference papers. We select three kinds of despeckling methods for comparisons and the first one is the traditional FANS [38], a fast variation of SAR-BM3D [35], that explores the handcrafted priors of sparseness and nonlocal property. The second type refers to DL-based approaches, including Speckle2Void [42], SAR-CNN [39], SAR-CAM [46], and Trans-SAR [48], and the last one is that with diffusion models, where only SAR-DDPM [72] is considered. Notice that the

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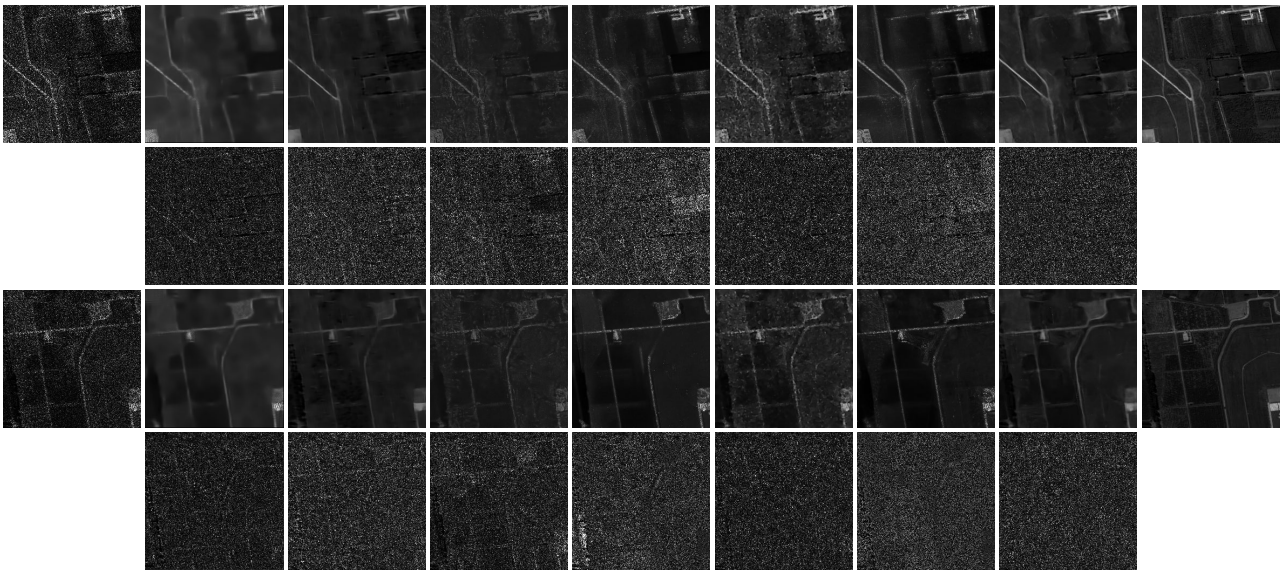


Fig. 1. Despeckling results of synthetic SAR images S1 and S2. (Top to Bottom) The despeckled images for S1, the relative ratio images for S1, the despeckled images for S2, and the relative ratio images for S2. SI: speckled image.(a) SI. (b) FANS. (c) Speckle2Void. (d) SAR-CNN. (e) SAR-CAM. (f) Trans-SAR. (g) SAR-DDPM. (h) SAR-DPS. (i) Ground truth.

unsupervised DDPM (U-DDPM) has close performance to the supervised ones [73], so U-DDPM is not tested for comparison. Similarly, the other methods, such as ID-CNN [40], SAR-ON [47], CNN-NLM [87] to name a few, are not considered as they have been illustrated to be inferior to SAR-DDPM [72].

5) Evaluation Metrics: To quantitatively evaluate the performance of the methods above, we leverage peak signal-to-noise ratio (PSNR) and structural similarity index (SSIM) to evaluate the despeckling results on synthetic SAR images, while ENL [84] and edge preservation index (EPI) [88] are used for the assessment of despeckling on real SAR images. Besides, we also compute the following metrics for both the synthetic SAR images and the real SAR images, namely: 1) MOI: the mean of the image; 2) MOR: the mean of the ratio image; 3) KL: the Kullback–Leibler divergence between statistical distribution of the ratio image and the Gamma distribution; and 4) M-Index [89].

6) Implementation Details: For SAR-DPS in Algorithm 1, we set $\gamma = 0.9$ and the other parameters for diffusion models follow the suggested ones in pretrained model [86]. Specifically, we set $T = 1000$ and choose the descending time stepping from $t = T$ to $t = 1$ with a uniformly spaced time-step schedule. Moreover, we adopt linear schedule for β_t from 0.0001 to 0.02 for 1000 timesteps, and $\{\widetilde{\sigma}_t\}_{t=1}^{T}$ are learned as in [86]. Besides, a small positive number $c > 0$ is added to $\hat{\mathbf{x}}_0 \odot \hat{\mathbf{x}}_0$ in (29) to avoid the zero elements of $\hat{\mathbf{x}}_0$, and at this point we set $c = 0.1$. All of the algorithms in experiments are performed in a single NVIDIA Quadro RTX6000 GPU with 24G video memory, and the code framework is PyTorch 1.13.1 based on Python 3.10.9.

B. Experiments on Synthetic SAR Images

Despeckling results of four synthetic SAR images, labeled S1, S2, S3, and S4, respectively, are presented in Figs. 1 and 2, where the scenes in images S1 and S2 are simple and that in images S3 and S4 are relatively complex. First, it can be observed that the proposed method SAR-DPS outperforms the compared methods for its good balance between the speckle suppression and detail preservation. Specifically, FANS can remove the speckle sufficiently but has the cost of detail loss. The DL-based methods either fall in the trouble of oversmoothing or have poor noise suppression. For example, SAR-CAM usually loses the details of the scenes while Trans-SAR cannot remove the noise clearly, especially for complex scenes. In visual effect, only the Speckle2Void and SAR-DDPM can offer similar despeckling results to SAR-DPS. Even so, SAR-DDPM needs supervised training and the difference in data distribution between the training samples and the testing samples usually results in performance loss of despeckling. Besides, we can observe that some small targets have been completely lost by the speckle, such as the aircraft on the bottom-right corner of the ground truth of S3 in Fig. 2. At this point, almost all of the methods are difficult to obtain the structure of the aircraft. For example, SAR-CNN and Trans-SAR only recover a target mixed with the aircraft and random speckle patterns, while SAR-DDPM even generates fake texture features due to weak measurement guidance and strong prior constraint in diffusion model. In comparison, the proposed SAR-DPS can avoid it by adopting the guidance approach of posterior sampling, leading to a good balance between measurement guidance and prior constraint. The corresponding ratio images in Figs. 1 and 2 also verify the statements above, and the ratio images for SAR-DPS are more uniform and have less structures than that for the compared methods.

The quantitative evaluation of the results in Figs. 1 and 2 is given in Table I, where we also present the ideal values of the evaluation metrics. We can see that the SAR-DPS has the optimal values for most of the metrics, which indicates that SAR-DPS not only has good noise suppression ability, but also preserves the details and general intensity values.

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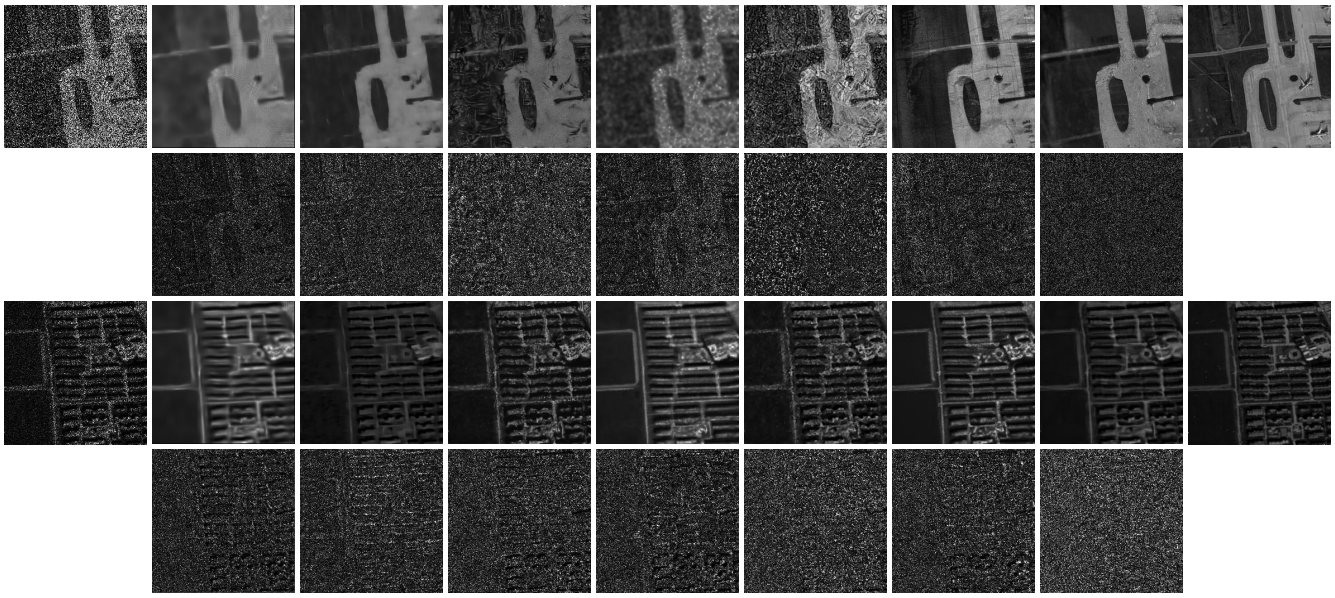


Fig. 2. Despeckling results of synthetic SAR images S3 and S4. (Top to Bottom) The despeckled images for S3, the relative ratio images for S3, the despeckled images for S4, and the relative ratio images for S4. SI: speckled image. (a) SI. (b) FANS. (c) Speckle2Void. (d) SAR-CNN. (e) SAR-CAM. (f) Trans-SAR. (g) SAR-DDPM. (h) SAR-DPS. (i) Ground truth.

TABLE I
EVALUATION OF DESPECKLING ON SYNTHETIC SAR IMAGES

Image	Metric	Ideal value	SAR-BMD	Speckle2Void	SAR-CNN	SAR-CAM	Trans-SAR	SAR-DDPM	SAR-DPS (ours)
S1	PSNR (dB)	Inf	24.38	27.31	25.42	25.71	24.06	26.01	**28.67**
	SSIM	1	0.89	0.90	0.91	0.89	0.88	0.90	**0.93**
	MOI	34	45	31	32	30	42	30	**35**
	MOR	1	0.75	1.07	1.05	1.12	0.78	1.11	**0.96**
	KL	0	0.12	0.04	0.06	0.10	0.14	0.07	**0.03**
	M-Index	0	11.32	8.87	9.12	11.78	15.62	12.92	**6.17**
S2	PSNR (dB)	Inf	26.28	28.91	27.15	27.39	26.73	29.13	**30.49**
	SSIM	1	0.88	0.92	0.90	0.88	0.87	0.91	**0.94**
	MOI	28	35	25	31	26	33	22	**27**
	MOR	1	0.76	**1.04**	0.86	1.07	0.79	1.19	0.95
	KL	0	0.16	0.07	0.13	0.11	0.17	0.16	**0.05**
	M-Index	0	12.91	9.49	11.24	10.51	14.71	12.24	**6.77**
S3	PSNR (dB)	Inf	21.53	**25.55**	20.18	21.17	18.02	23.19	25.28
	SSIM	1	0.87	0.90	0.85	0.88	0.84	0.91	**0.92**
	MOI	69	85	72	61	68	82	77	**70**
	MOR	1	0.78	0.91	1.22	**1.05**	0.83	0.89	0.94
	KL	0	0.18	0.08	0.16	0.05	0.12	0.08	**0.04**
	M-Index	0	19.17	8.53	13.96	9.42	14.73	11.67	**7.25**
S4	PSNR (dB)	Inf	20.06	24.41	22.97	19.76	23.07	23.77	**25.87**
	SSIM	1	0.79	0.91	0.89	0.80	0.88	0.90	**0.93**
	MOI	31	45	28	37	41	36	40	**30**
	MOR	1	0.85	1.11	0.92	0.89	0.91	0.87	**1.05**
	KL	0	0.17	0.09	0.12	0.15	0.11	0.14	**0.06**
	M-Index	0	21.22	9.78	12.76	13.56	15.23	14.94	**8.46**

The success of SAR-DPS stems from two aspects and the first one is the pretraining model, which not only solves the difficulty of collecting training samples and high computation complexity of training, but also fuses the priors from the optical natural images (i.e., the ImageNet dataset) and the measurement information from the speckled images. The other factor lies in the exact exploration of the statistical model of speckle in posterior sampling, where no transforms are introduced to approximate the speckle distribution in transform domain by some simple distributions (i.e., Gaussian distribution).

C. Experiments on TerraSAR-X Images

TerraSAR-X images, labeled as T1, T2, T3, T4, T5, and T6, respectively, are from the spaceborne X-band SAR sensor with the staring spot light mode and all of them are single-look complex (SLC) data over the scenes of harbor, farmland, city, and so on. In this section, we only employ the T1 and T2 to validate the superiority of the proposed method, while we will utilize from T3 to T6 to present the robustness of the proposed method in Section V-G.

Despeckling results of TerraSAR images T1 and T2 are shown in Fig. 3, and we can easily observe that the details of

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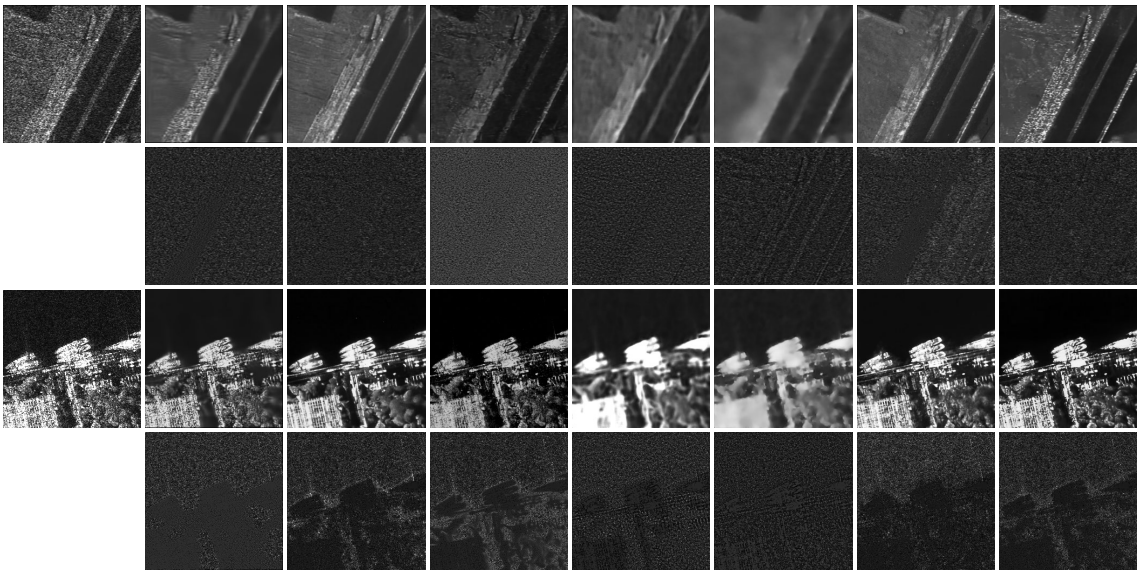


Fig. 3. Despeckling results of TerraSAR-X images T1 and T2. (Top to Bottom) The despeckled images for T1, the relative ratio images for T1, the despeckled images for T2, and the relative ratio images for T2. (a) SAR image. (b) FANS. (c) Speckle2Void. (d) SAR-CNN. (e) SAR-CAM. (f) Trans-SAR. (g) SAR-DDPM. (h) SAR-DPS.

TABLE II
EVALUATION OF DESPECKLING ON TeeraSAR-X IMAGES

Image	Metric	Ideal value	SAR-BMD	Speckle2Void	SAR-CNN	SAR-CAM	Trans-SAR	SAR-DDPM	SAR-DPS (ours)
T1	ENL	Inf	34.83	35.78	24.26	38.67	**41.85**	36.94	39.93
	EPI	1	0.16	0.21	0.19	0.17	0.07	0.23	**0.37**
	MOI	66	**65**	70	44	68	73	71	64
	MOR	1	0.92	**0.97**	1.49	0.94	0.88	0.92	1.04
	KL	0	0.08	0.07	0.23	0.10	0.12	0.09	**0.06**
	M-Index	0	7.42	6.34	14.06	9.69	11.25	7.58	**5.71**
T2	ENL	Inf	25.64	27.83	23.15	28.62	**30.23**	26.07	29.26
	EPI	1	0.51	0.47	0.5	0.28	0.24	0.48	**0.53**
	MOI	76	66	69	64	89	85	70	**71**
	MOR	1	1.11	1.07	1.13	0.93	0.86	1.08	**1.05**
	KL	0	0.13	0.11	0.15	0.16	0.21	0.09	**0.07**
	M-Index	0	13.48	9.54	14.46	17.19	21.84	10.96	**8.79**

the scenes are not well preserved by SAR-CAM and Trans-SAR. And then, it is also noted that the despeckled images by SAR-CNN become dark and the speckle still remains. The other methods can achieve good balance between speckle suppression and detail preservation, while Speckle2Viod can preserve spatial details well and SAR-DDPM and SAR-DPS seem oversmoothed for T1. However, there are more details in the bottom-right area of the T2 image for SAR-DPS than that for FANS, Speckle2Void, and SAR-DDPM. With respect to the ratio images, SAR-DPS does not show any structures in the ratio image of T1, but we can notice clear details in the ratio image of T2 for SAR-DPS. This may result from the original SAR image of T2, since its pixels with high intensity seemly have been truncated and the corresponding speckle disappears. In fact, this phenomenon can be observed more or less in all of the ratio images of T2, and it can also be noticed in some of the following results.

The quantitative evaluation results for Fig. 3 are shown in Table II. For ENL, Trans-SAR achieves the highest values for both T1 and T2, and SAR-DPS is the second best, which is in agreement with the visual effects. Besides, SAR-DPS outperforms the compared methods in terms of EPI, indicating that the proposed method has the good ability of structure preservation. Generally speaking, SAR-DPS obtains the optimal values for most of the metrics and the suboptimal values for the other ones.

D. Experiments on Gaofen-3 Images

Gaofen-3 images, labeled GF1 and GF2, respectively, are from spaceborne C-band SAR sensor with the mode of spot light, and both of them are HH polarization, with the resolution of 1 $\times$ 1 m for range and azimuth directions, and SLC data with the scenes of architectures and roads.

Despeckling results of Gaofen-3 images GF1 and GF2 are presented in Fig. 4, from which we can see that the edges and structures in original SAR images are rich enough and so they are challenging for despeckling. Specifically, the despeckled images are obviously oversmoothed by SAR-CAM and Trans-SAR, while Speckle2Void and SAR-DDPM produce the ratio images with clear structures and details, validating their inferiority of edge and spatial preservation. For the ration

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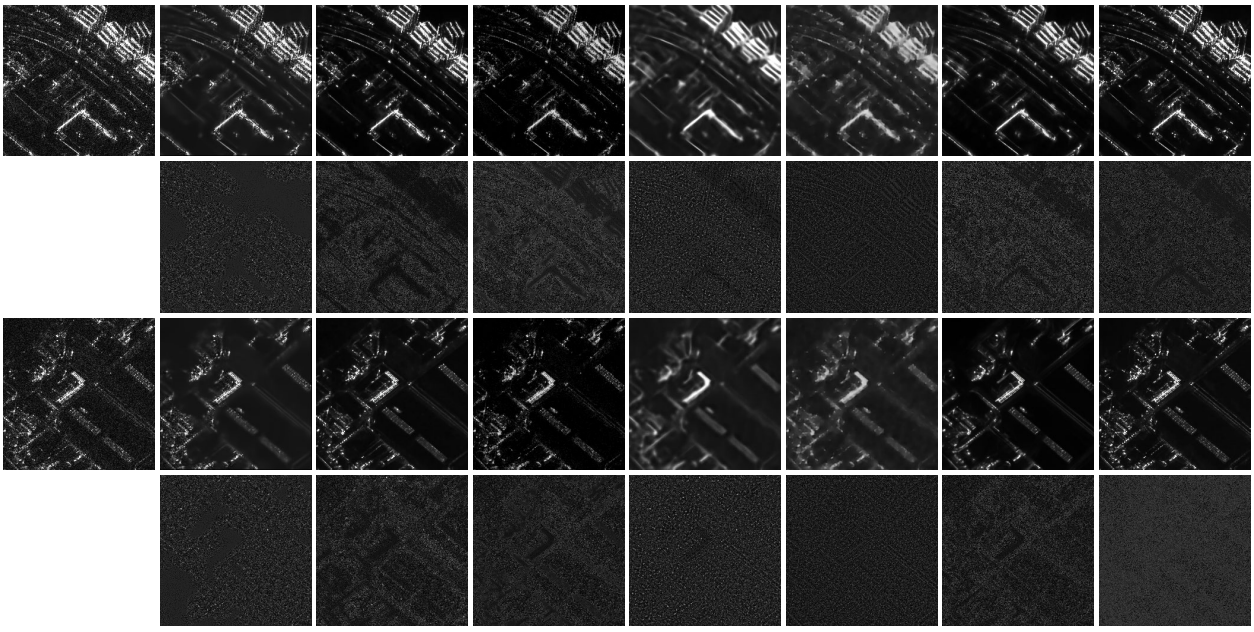


Fig. 4. Despeckling results of Gaofen-3 images GF1 and GF2. (Top to Bottom) The despeckled images for GF1, the relative ratio images for GF1, the despeckled images for GF2, and the relative ratio images for GF2. (a) SAR image. (b) FANS. (c) Speckle2Void. (d) SAR-CNN. (e) SAR-CAM. (f) Trans-SAR. (g) SAR-DDPM. (h) SAR-DPS.

TABLE III
EVALUATION OF DESPECKLING ON GAOFEN-3 IMAGES

Image	Metric	Ideal value	SAR-BMD	Speckle2Void	SAR-CNN	SAR-CAM	Trans-SAR	SAR-DDPM	SAR-DPS (ours)
GF1	ENL	Inf	28.53	30.96	24.73	29.03	28.42	31.13	**32.27**
	EPI	1	0.45	0.57	0.55	0.43	0.36	0.48	**0.62**
	MOI	46	41	43	42	49	52	50	**44**
	MOR	1	1.09	1.07	1.08	**0.98**	0.83	0.96	1.03
	KL	0	0.12	0.08	0.11	0.06	0.15	0.07	**0.05**
	M-Index	0	12.34	8.44	15.71	10.59	18.92	9.63	**7.19**
GF2	ENL	Inf	43.11	37.15	32.94	46.37	45.79	40.61	**47.52**
	EPI	1	0.34	0.51	0.58	0.22	0.28	0.36	**0.60**
	MOI	35	33	32	30	**35**	39	31	33
	MOR	1	1.03	1.05	1.10	**0.99**	0.86	1.09	1.02
	KL	0	0.05	0.04	0.12	0.06	0.07	0.09	**0.03**
	M-Index	0	14.26	7.22	16.53	10.48	8.94	8.45	**6.51**

images of FANS and SAR-CNN, they have more details than that for SAR-DPS, and so SAR-DPS still outperforms the compared methods for Gaofen-3 data and performs well for different scenes. The statements above can also be verified by the quantitative assessment in Table III. For ENL and EPI, SAR-DPS achieves the highest values among these methods, indicating the best performance of speckle suppression and detail preservation. Meanwhile, for the GF1 image, SAR-DPS has the second-best values of the MOR, and so does it for the MOI and the MOR for the GF2 image. For the other metrics, SAR-DPS performs best, and so we can conclude that SAR-DPS is effective in suppressing the noise and preserving the SAR texture for Gaofen-3 data.

In Fig. 4, we also notice that some ratio images have inversion phenomenon, namely, homogeneous areas and strong scatters in SAR image correspond to strong texture and weak texture in ratio images, respectively. Let us take the ratio images of Speckle2Void in Fig. 4 for example. Generally speaking, the speckle in homogeneous areas is easy to be suppressed and so the ratio image has little or no texture, while despeckling is difficult for the strong scatters and so the ratio image often has strong texture. However, the ratio images for Speckle2Void exist a reversion phenomenon, which can be explained by the SAR image and the despeckling method. On the one hand, the original SAR images in Fig. 4 have large dynamic range, and the intensity of homogeneous areas (e.g., the background) almost reaches the lower bound and there are also many strong scatters whose intensity almost reaches the upper bound. On the other hand, we observe that the estimated intensities for homogeneous areas and strong scatters by Speckle2Void are almost the lower bound and the upper bound, respectively, namely, Speckle2Void is actually a biased estimation for the truth. Therefore, the ratio image for strong scatters almost reaches one, while that for homogeneous areas easily reach large value if the lower bound is too small. In this case, the ration image for strong scatters seems dark but that for homogeneous areas seems light, namely the reversion phenomenon appears in the ratio image. In a word, for the SAR

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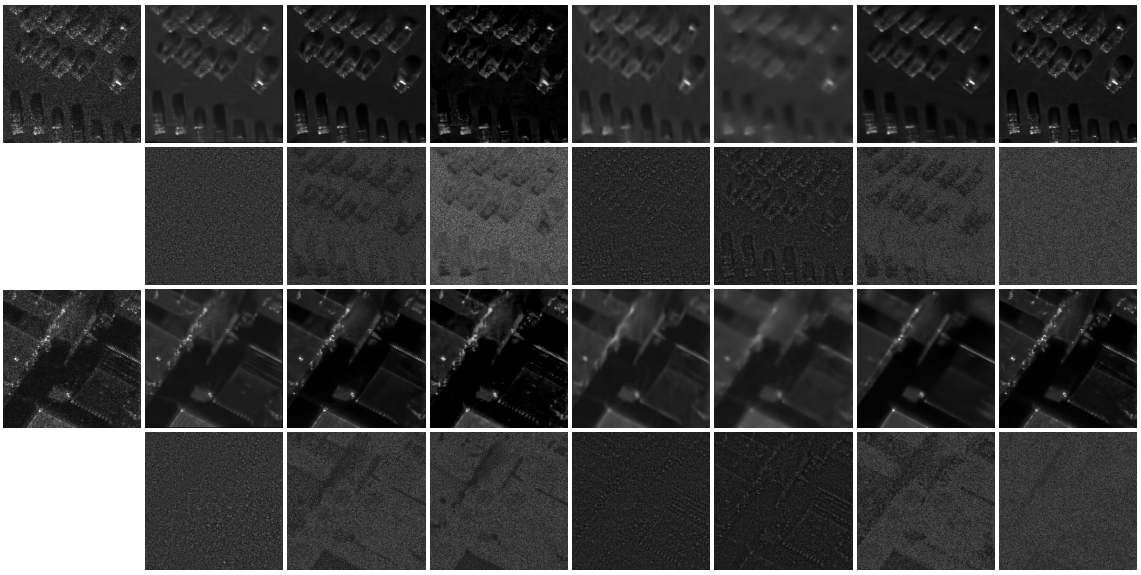


Fig. 5. Despeckling results of MiniSAR images MS1 and MS2. (Top to Bottom) The despeckled images for MS1, the relative ratio images for MS1, the despeckled images for MS2, and the relative ratio images for MS2. (a) SAR image. (b) FANS. (c) Speckle2Void. (d) SAR-CNN. (e) SAR-CAM. (f) Trans-SAR. (g) SAR-DDPM. (h) SAR-DPS.

images with large dynamic range, the version phenomenon in the ratio image can be avoided only if the despeckling method exhibits like an unbiased estimation.

E. Experiments on MiniSAR Images

To further examine the despeckling performance of the SAR-DPS, the MiniSAR open data with the file tags of MiniSAR20050519p0005image003 and MiniSAR20050519p0006image005 are used, where two subimages with the size of 256 $\times$ 256 are selected for experiments and they are labeled MS1 and MS2, respectively. MiniSAR is airborne Ku-band sensor, and the selected images are produced in the spot light mode and the spatial resolution is about 0.1 $\times$ 0.1 m for range and azimuth directions. For imaging scenes, there are vehicles and buildings for MS1 and MS2 images, respectively.

Despeckling results of MiniSAR images MS1 and MS2 are presented in Fig. 5. For the original image of MS1, it can be noticed that the shadow of the vehicles may cause interference for despeckling. Similarly, there are a series of points with small distances in the middle-right area of the original image of MS2. Both of them are challenging for despeckling, and most of the methods are difficult to well tackle these hurdles in Fig. 5. In fact, for the image of MS1, the methods of FANS, SAR-CAM, Trans-SAR, and SAR-DDPM do not recover the clear edges between the shadow and the background, while only Speckle2Void and SAR-DPS succeed. However, there are more structures in the ratio image for Speckel2Void than that for SAR-DPS. For the image of MS2, the aforementioned points can be only preserved by FANS, Speckle2Void, and SAR-DPS, and we can only observe a line instead of points in the despeckled images by the other methods. Even so, these points are also blurred by FANS. With respect to the ratio images, SAR-DPS has less structures and details than the compared methods, indicating its ability of speckle suppression and structure preservation.

The values in Table IV offer quantitative evaluation for the despeckling methods. More preciously, SAR-DPS has the optimal or suboptimal values for all the metrics, and so its performance is best among these methods. Speckle2Void and FANS also have preferable values of the metrics in general, which is consistent with the visual effects. Although some methods also achieve optimal values of some metrics, their limitations on the other metrics are obvious. For example, Trans-SAR has the highest ENL values for MS1 and MS2, but its EPI values are the lowest. In this case, the despeckling by Trans-SAR may not be accepted.

F. Experiments on RADARSAT-1 Images

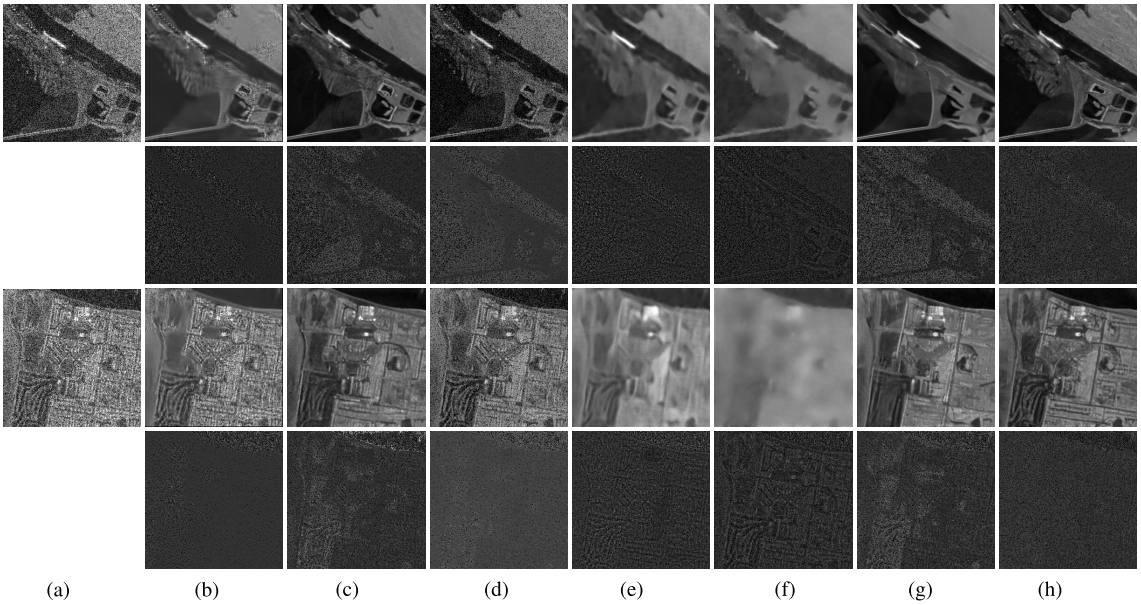
RADARSAT-1 images, labeled RD1 and RD2, respectively, are from the spaceborne C-band SAR sensor with the strip map mode and both of them are HH polarization. Their resolution is 8 $\times$ 8 m for the range and azimuth directions, and they are the SLC data with the scenes of harbor, village, and so on.

Despeckling results of RADARSAT-1 images RD1 and RD2 are shown in Fig. 6. Notice that the original SAR images have low resolutions and the scenes are relatively complex, and so it is difficult to keep the balance between speckle suppression and detail preservation. Specifically, FANS and SAR-CNN cannot remove the noise completely, since there are lots of noise left in their despeckled images. Moreover, SAR-CAM and Trans-SAR obviously result in oversmoothing effects, and the structures and details of the objects in the scenes have been blurred. Although the visual effects of the despeckled images by Speckled2Void and SAR-DDPM can be accepted, their ratio images contain lots of structures. In summary, SAR-DPS has the best performance in terms of the despeckled image and the ratio image.

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TABLE IV
EVALUATION OF DESPECKLING ON MiniSAR IMAGES

Image	Metric	Ideal value	SAR-BMD	Speckle2Void	SAR-CNN	SAR-CAM	Trans-SAR	SAR-DDPM	SAR-DPS (ours)
MS1	ENL	Inf	34.13	37.38	31.95	41.25	**44.59**	38.13	43.56
	EPI	1	0.11	0.15	0.10	0.13	0.09	0.12	**0.16**
	MOI	45	43	40	37	**46**	50	41	43
	MOR	1	1.04	1.07	1.13	**0.98**	0.89	1.06	1.03
	KL	0	0.06	0.09	1.11	0.07	0.12	0.10	**0.04**
	M-Index	0	9.39	7.26	13.37	10.24	14.63	7.18	**6.25**
MS2	ENL	Inf	35.20	36.92	32.75	35.62	**39.71**	34.79	**37.81**
	EPI	1	0.28	0.41	0.36	0.22	0.13	0.31	**0.42**
	MOI	38	**36**	32	29	**40**	42	33	35
	MOR	1	**1.03**	1.09	1.15	0.94	0.88	1.07	1.04
	KL	0	0.07	0.10	0.12	0.05	0.14	0.09	**0.04**
	M-Index	0	10.18	9.26	14.28	12.72	16.37	9.54	**7.63**



(a) (b) (c) (d) (e) (f) (g) (h)

Fig. 6. Despeckling results of RADARSAT-1 images RD1 and RD2. (Top to Bottom) The despeckled images for RD1, the relative ratio images for RD1, the despeckled images for RD2, and the relative ratio images for RD2. (a) SAR image. (b) FANS. (c) Speckle2Void. (d) SAR-CNN. (e) SAR-CAM. (f) Trans-SAR. (g) SAR-DDPM. (h) SAR-DPS.

TABLE V
EVALUATION OF DESPECKLING ON RADARSAT-1 IMAGES

Image	Metric	Ideal value	SAR-BMD	Speckle2Void	SAR-CNN	SAR-CAM	Trans-SAR	SAR-DDPM	SAR-DPS (ours)
RD1	ENL	Inf	39.74	40.23	5.57	40.99	**45.45**	38.67	41.72
	EPI	1	0.26	0.31	**0.67**	0.17	0.14	0.32	0.34
	MOI	76	**78**	69	63	80	86	70	73
	MOR	1	0.95	1.11	1.16	0.96	0.12	0.09	**1.03**
	KL	0	0.09	0.08	0.14	0.06	0.12	0.09	**0.05**
	M-Index	0	13.18	10.76	20.51	18.63	15.55	11.84	**9.01**
RD2	ENL	Inf	28.10	29.66	9.94	28.92	**55.47**	27.97	30.12
	EPI	1	0.22	0.27	**0.69**	0.11	0.07	0.20	0.29
	MOI	110	105	108	103	119	123	100	**109**
	MOR	1	1.06	1.05	1.08	0.93	0.88	1.13	**1.04**
	KL	0	0.12	0.09	0.08	0.14	0.16	0.06	**0.04**
	M-Index	0	14.92	12.75	15.18	17.54	16.67	14.21	**10.82**

Besides, we also notice in Fig. 6 that many black spots appear in the despeckled images of FANS and SAR-CNN. First, we observe that there are also many black spots in original SAR images, which may result from their small quantization bits. In other words, the speckle with low intensity may be digitalized into zero (i.e., the black spots) if the quantization bits are small. Second, FANS and SAR-CNN cannot remove these black spots in despeckling because of failing to recognize them as speckle, and so we can conclude that they do not suit well for suppressing the speckle for SAR images with small quantization bits.

The quantitative evaluation of Fig. 6 is shown in Table V, from which we can have the similar conclusion to the analysis above. More preciously, the compared methods may have the optimal values for some metrics, but they usually suffer from the limitations of the other metrics. For instance, SAR-CNN

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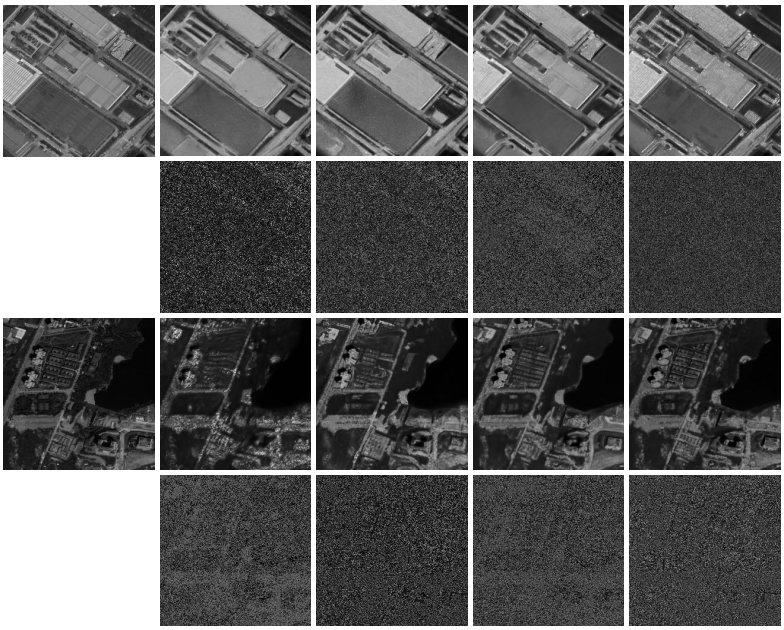


Fig. 7. Despeckling results of synthetic SAR images S5 and S6 with different speckle levels. (Top to Bottom) The despeckled images for S5, the relative ratio images for S5, the despeckled images for S6, and the relative ratio images for S6.(a) Ground truth. (b) $L = 1$. (c) $L = 2$. (d) $L = 3$. (e) $L = 4$.

TABLE VI
EVALUATION OF DESPECKLING ON SYNTHETIC SAR IMAGES S5 AND S6 WITH DIFFERENT SPECKLE LEVELS

Image	Metric	Ideal value	$L = 1$	$L = 2$	$L = 3$	$L = 4$
S5	PSNR (dB)	Inf	21.42	22.34	24.77	26.58
	SSIM	1	0.90	0.91	0.93	0.95
	MOI	63	65	64	62	63
	MOR	1	0.96	0.97	1.04	1.02
	KL	0	0.06	0.05	0.03	0.02
	M-Index	0	9.32	7.68	3.52	2.46
S6	PSNR (dB)	Inf	22.15	23.94	25.11	27.65
	SSIM	1	0.91	0.92	0.93	0.96
	MOI	38	41	40	39	39
	MOR	1	0.96	0.95	0.97	0.98
	KL	0	0.05	0.03	0.03	0.02
	M-Index	0	8.26	7.45	5.38	4.95

obtains the highest EPI values for both RD1 and RD2, but its ENL values are the lowest. Totally speaking, SAR-DPS achieves the optimal values for most of the metrics, and more importantly it does not have any limitations from the other metrics. Therefore, we can conclude that SAR-DPS has the best performance for RADARSAT-1 images.

G. Robustness Analysis

To analyze the robustness of the proposed method, we first employ the synthetic SAR images S5 and S6 to examine the performance of SAR-DPS under the conditions of different speckle levels. And then, SAR-DPS is tested on the TerraSAR-X images T3 and T4 with different ISs of diffusion model. Finally, TerraSAR images T5 and T6 are used to validate the performance of SAR-DPS with different values of the algorithm parameter γ. These SAR images have been introduced in Sections V-B and V-C, and so we do not show their details anymore.

1) *Speckle Level*: Despeckling results of synthetic SAR images S5 and S6 with different speckle levels as well as the corresponding quantitative evaluation are presented in Fig. 7 and Table VI, respectively. We can see that the performance of SAR-DPS becomes better and better as the speckle intensity gets lower and lower. Specifically, although SAR-DPS has good speckle suppression performance for different speckle levels, we can see that the details in the despeckled images become more and more as L becomes larger. Correspondingly, the structures and details left in the ratio images also become less and less. More preciously, as L gets larger, the values of PSNR and SSIM become larger, indicating the ability of SAR-DPS for keeping the balance of speckle suppression and detail preservation becomes stronger. Meanwhile, the values of KL and M-Index become smaller as L gets larger, but the values of MOI and MOR receive less influence than the other metrics from different speckle levels.

2) *Initial Sample*: Notice that the reverse denoising process of DMs in SAR-DPS always starts from a random noise $\mathbf{x}_T$

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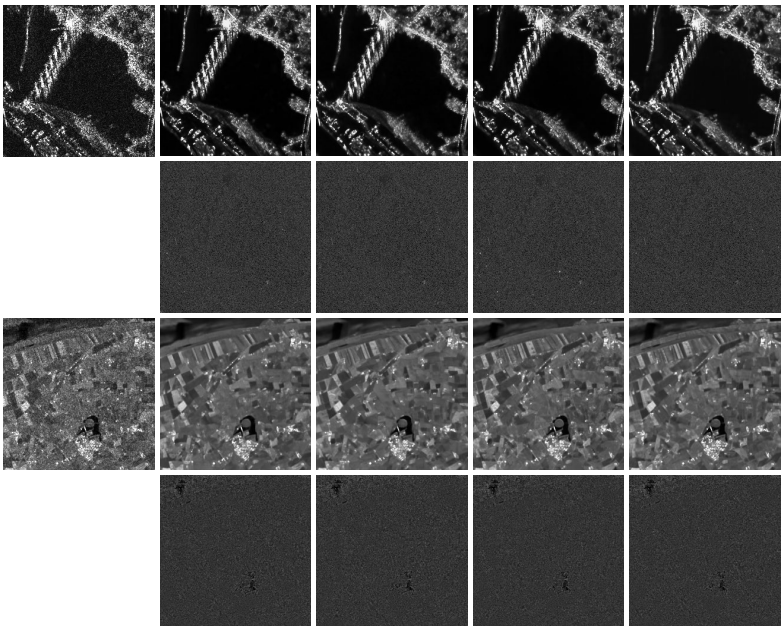


Fig. 8. Despeckling results of TerraSAR-X images T3 and T4 with different ISs. (Top to Bottom) The despeckled images for T3, the relative ratio images for T3, the despeckled images for T4, and the relative ratio images for T4. (a) SAR image. (b) IS #1. (c) IS #2. (d) IS #3. (e) IS #4.

TABLE VII
EVALUATION OF DESPECKLING ON TERRASAR-X IMAGES T3 AND T4 WITH DIFFERENT ISS

Image	Metric	Ideal value	IS #1	IS #2	IS #3	IS #4
T3	ENL	Inf	36.97	36.83	36.68	36.72
	EPI	1	0.56	0.57	0.55	0.57
	MOI	55	53	54	52	53
	MOR	1	1.03	1.02	1.04	1.02
	KL	0	0.05	0.04	0.05	0.06
	M-Index	0	6.35	6.29	6.31	6.44
T4	ENL	Inf	28.35	28.42	28.34	28.39
	EPI	1	0.51	0.53	0.52	0.51
	MOI	93	92	90	91	93
	MOR	1	1.02	1.03	1.04	1.02
	KL	0	0.03	0.04	0.03	0.05
	M-Index	0	7.43	7.58	7.52	7.49

sampled from $\mathcal{N}(\mathbf{0}, \mathbf{I})$, and so we intend to perform the reverse denoising process of DMs with different ISs on TerraSAR-X images T3 and T4. Namely, we aim to test SAR-DPS under different ISs from $\mathcal{N}(\mathbf{0}, \mathbf{I})$ to validate its robustness. The despeckling results are presented in Fig. 8, and the corresponding quantitative assessment is shown in Table VII. We can note that, for T3 and T4, the relative variations of the performance of SAR-DPS with respect to random initialization are small in terms of the visual effects and the values of the metrics, although T3 and T4 have different scenes. We also notice that there are some structures left in the ratio images, which may result from the truncation of the highest pixels and the lowest pixels in original images.

3) Algorithm parameter γ. At this point, we test SAR-DPS with different values of the algorithm parameter γ, and this parameter actually stands for the guidance intensity. We set the γ values from 0.5 to 1.3 with the uniform space of 0.2, and the despeckling results are shown in Fig. 9. In visual effects, it is hard to observe the variations of the despeckled images, and the difference of the statistical distributions of ratio images is also tiny. Besides, we can see the slight truncation phenomenon in the ratio images, and they are also stable with respect to the algorithm parameter γ. Different from the visual effects in Fig. 9, the quantitative evaluation in Table VIII can provide the variations of SAR-DPS with respect to different values of γ. Specifically, as the γ value grows up, the ENL value becomes smaller while the EPI value gets larger. This is because the large value of γ means high guidance intensity, namely, the weight of the data-fidelity term in the model of solving the despeckling inverse problem is large. In this case, more details can be preserved, and so the above change of ENL and EPI can be observed. The other metrics do not change greatly as the γ value becomes larger, so the performance of SAR-DPS is robust to the parameter γ in general and we set $\gamma = 0.9$ in the experiments above to balance the speckle suppression and detail preservation.

Finally, we aim to conclude the experiments by some remarks: 1) the proposed method SAR-DPS not only works

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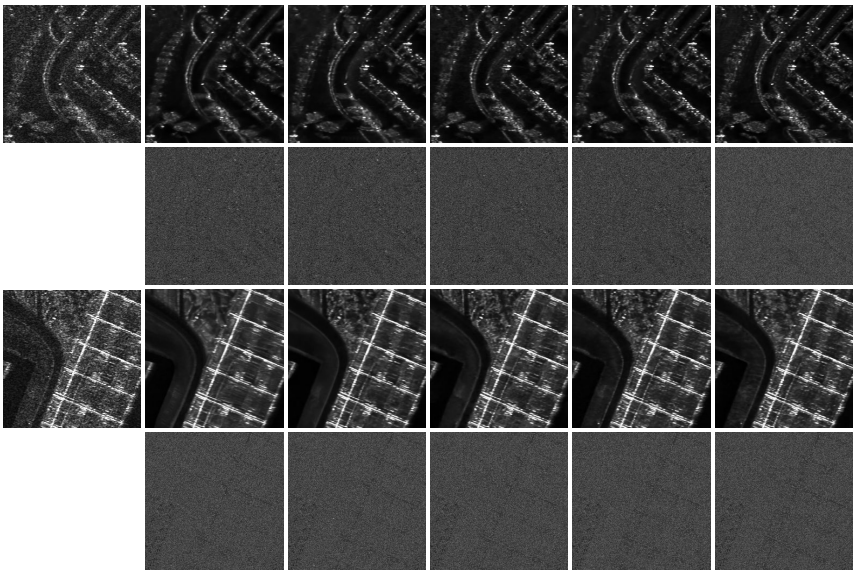


Fig. 9. Despeckling results of TerraSAR-X images T5 and T6 with different values of the parameter γ. (Top to Bottom) The despeckled images for T5, the relative ratio images for T5, the despeckled images for T6, and the relative ratio images for T6. (a) SAR image. (b) $\gamma = 0.5$. (c) $\gamma = 0.7$. (d) $\gamma = 0.9$. (e) $\gamma = 1.1$. (f) $\gamma = 1.3$.

TABLE VIII
EVALUATION OF DESPECKLING ON TeeraSAR-X IMAGES T5 AND T6 WITH DIFFERENT VALUES OF THE PARAMETER γ

Image	Metric	Ideal value	$\gamma = 0.5$	$\gamma = 0.7$	$\gamma = 0.9$	$\gamma = 1.1$	$\gamma = 1.3$
T5	ENL	Inf	38.42	38.03	37.86	36.49	35.82
	EPI	1	0.48	0.49	0.51	0.52	0.54
	MOI	47	44	45	45	46	45
	MOR	1	1.04	1.05	1.03	1.04	1.03
	KL	0	0.05	0.04	0.04	0.06	0.05
	M-Index	0	7.43	7.18	7.29	7.61	7.37
T6	ENL	Inf	32.47	32.02	31.73	30.96	30.19
	EPI	1	0.26	0.27	0.27	0.29	0.30
	MOI	66	64	63	66	65	65
	MOR	1	1.05	1.04	1.05	1.03	1.04
	KL	0	0.06	0.07	0.06	0.08	0.07
	M-Index	0	8.17	8.03	7.94	8.12	7.89

well for synthetic SAR images, but also suits for various real SAR images with different bands, different resolutions, and different scenes; 2) the image scenes usually have great influence on the performance of SAR-DPS as well as the compared methods. If the scenes are simple, we can make use of SAR-DPS to reach a good tradeoff between speckle suppression and detail preservation, otherwise the despeckling effect would decrease. 3) SAR-DPS can be further improved via simple skills. For example, the popular techniques, such as circle spinning [72] and averaging [73], can be utilized to enhance the image quality for SAR-DPS. 4) As stated in [72], the methods based on diffusion models for despeckling are seriously limited by the high computation complexity. For example, SAR-DPS takes more than 1 min to implement inference and generation of an image with the size of 256 $\times$ 256, while Trans-SAR and SAR-CAM take no more than 1 s for the same task. This limitation mainly stems from diffusion models themselves, and several standard strategies, such as early exiting [90], shortcut sampling [91], and GPU-aware optimization [92], have been proposed to accelerate the diffusion models. Specifically, we can also make use of dimension reduction skills to decrease computation cost for large-size SAR images, and these skills include but are not limited to dividing SAR images into sub-blocks and transforming them into latent space [93]. 5) For a fair comparison, we employ the same training sample set to retrain the networks of the DL-based methods instead of directly utilizing the trained networks provided by their authors. In this case, the networks may not be trained with good convergence, leading to insufficient despeckling in some experimental results. Even so, we do not intend to augment the training samples, since we believe that the adaptation to a limited training set is also an important measure for the DL methods.

VI. CONCLUSION

We propose a novel SAR-DPS for SAR despeckling, achieving a good tradeoff between speckle suppression and detail preservation via exploring the deep generative priors by diffusion models. The kernel of SAR-DPS lies in the calculation of likelihood score function, where the reverse transition distribution is first estimated and then the multiplicative noise

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model is utilized to compute the measurement distribution with the condition of the ground truth. Subsequently, the algorithm of SAR-DPS is summarized and its superiority is also analyzed theoretically. In the experiments, the performance of SAR-DPS is validated on synthetic and real SAR images, and the results indicate that the proposed method outperforms the compared approaches in terms of visual effects and quantitative evaluation. Besides, its robustness with respect to speckle intensity, ISs of diffusion model, and the algorithm parameter is also illustrated.

SAR-DPS provides a general framework of diffusion posterior sampling for SAR despeckling, since various traditional skills, such as optimizing the network structure, increasing training samples, modifying diffusion schemes, and so on, can be explored to further improve its capability. More importantly, we think there are three research directions in future works. The first work lies in the refinement of the reverse transition distribution, which is currently modeled as one-point distribution (i.e., the variance is zero). In fact, this assumption usually does not hold in practical applications, although its computation is simple. The second direction is the acceleration of diffusion model for SAR-DPS, which aims to reduce the computation complexity while keeping the processing precision. The final direction is the generation of SAR-DPS from single-channel SAR images to multichannel ones, such as POLSAR data. Diffusion model is also good at fusing channel information and so it is possible to apply to the correlated datasets.

APPENDIX
PROOF OF THE EQUATION IN (11)

Assume that the forward diffusion process in (5) [or equivalently in (3)] and the reverse denoising process in (7) are the discrete versions of score-based DMs in (9) and (10), respectively. According to (6), we have

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) = -\frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{1-\bar{\alpha}_t}. \tag{31}$$

Associated with (5), we can further have

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) = -\frac{\boldsymbol{\epsilon}}{\sqrt{1-\bar{\alpha}_t}}. \tag{32}$$

Notice that the noise estimation network $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ in DDPM and the score matching network $\mathbf{s}_\theta(\mathbf{x}_t, t)$ in score-based DMs are used to approximate the noise and the score function, respectively, in the training of the networks, so it holds that

$$\begin{cases} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \approx \boldsymbol{\epsilon} \\ \mathbf{s}_\theta(\mathbf{x}_t, t) \approx \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t). \end{cases} \tag{33}$$

Substituting the equation above into (32), we can readily conclude the statement in (11).

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